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- Show your work for all problems on the pages given or on scrap paper.
- Please write your name on **every page** that you use to show your work.
- You may use the front and back of the paper.
- Clearly **label** your work with the problem number you are working on.
- **Record which problems you worked** on each page (front and back).
- When you have completed the exam, count the number of pages you used, then **label each page** in order of the problem number.
- You will **staple** your pages together with **this page on top** when you complete your exam.

INSTRUCTOR USE ONLY

Problem number	Point Adjustment	Out of	Notes
1		1	
2		1.2	
3		1	
4		1.2	
5		1.2	
6		1.2	
7		1	
8		1	
9		1.2	
Total adjustment		10	ALEKS SCORE: Adjusted total score:

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Formulas for Retention Exam

$$(a - b)^3 = (a - b)(a^2 + ab + b^2)$$

$$(a + b)^3 = (a^3 + 3a^2b + 3ab^2 + b^3)$$

$$(a^2 - b^2) = (a - b)(a + b)$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x)-f(a)}{x-a}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} \quad AV_{[a,b]} = \frac{f(b)-f(a)}{b-a}$$

Point-slope form of a line with slope m , through

$$(x_1, y_1) : y = m(x - x_1) + y_1$$

Slope of a line with slope through

$$(x_1, y_1) \text{ and } (x_2, y_2); m = \frac{y_2 - y_1}{x_2 - x_1}$$

Student Name: _____

Class Name: **Calculus 1 - fall 2025**Number of Questions: **9**Instructor Name: XXXXXXXXXX**Question 1 of 9**

Complete the parts below to determine the apparent value of the following limit.

$$\lim_{x \rightarrow 5} \frac{7(x-5)}{x^2-25}$$

$x = 4.999$
 $\lim_{x \rightarrow 5} \frac{7(4.999-5)}{(4.999)^2-25} = \frac{-0.007}{-0.009999} = 0.700070007$
 $\lim_{x \rightarrow 5} \frac{7(5.001-5)}{(5.001)^2-25} = \frac{0.007}{0.010001} = 0.699930007$
0.7001

$$\lim_{x \rightarrow 5} \frac{7(5.001-5)}{(5.001)^2-25} = \frac{0.007}{0.010001} = 0.699930007$$

0.6999

(a) Fill in the blanks. Do not round intermediate computations, and round your answers to 4 decimal places where applicable.

x	4	4.5	4.95	4.999	$\rightarrow 5 \leftarrow$	5.001	5.05	5.5	6
$\frac{7(x-5)}{x^2-25}$	0.7778	0.7368	0.7035	<u>0.7001</u>	$\rightarrow ? \leftarrow$	<u>0.6999</u>	0.6965	0.6667	0.6364

(b) Use the values from part (a) to fill in the apparent value of the following limit.

$$\lim_{x \rightarrow 5} \frac{7(x-5)}{x^2-25} = \underline{0.7}$$

Question 2 of 9

Let $f(x) = 4x^2$. $f(2) = 4(2)^2 = 16$

(a) For $h \neq 0$, find the difference quotient $\frac{f(2+h)-f(2)}{h}$.Simplify your answer so that the resulting expression is defined at $h=0$.

$$\frac{f(2+h)-f(2)}{h} = \underline{16+4h}$$

(b) Find $f'(2)$.

$$f'(2) = \underline{16+4(0)} = 16$$

$$\lim_{h \rightarrow 0} \frac{4(2+h)^2 - 16}{h} = \lim_{h \rightarrow 0} \frac{4(2+h)(2+h) - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{4(4+4h+h^2) - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{16 + 16h + 4h^2 - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{16h + 4h^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{h(16+4h)}{h}$$

$$\lim_{h \rightarrow 0} 16+4h$$

Question 3 of 9

Find the following limit.

If the limit does not exist, write "Does Not Exist."

$$\lim_{x \rightarrow 1} \frac{x^2-2x+7}{x-4} = \frac{(1)^2-2(1)+7}{1-4} = \frac{1-2+7}{-3} = \frac{6}{-3} = \underline{-2}$$

Question 4 of 9

Let the function f be defined as follows.

$$f(x) = 2x^2 + x \quad f'(x) = \lim_{h \rightarrow 0} \frac{2(x+h)^2 + (x+h) - (2x^2 + x)}{h}$$

Answer the following.

- (a) For $h \neq 0$, find the difference quotient $\frac{f(x+h) - f(x)}{h}$.

Simplify your answer so that the resulting expression is defined at $h = 0$.

$$\frac{f(x+h) - f(x)}{h} = \underline{4x + 2h}$$

- (b) Find $f'(x)$.

$$f'(x) = \underline{4x}$$

$$\lim_{h \rightarrow 0} \frac{2(x+h)(x+h) + (x+h) - (2x^2 + x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) + (x+h) - (2x^2 + x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + x + h - 2x^2 - x}{h}$$

$$\lim_{h \rightarrow 0} \frac{4xh + 2h^2 + h}{h} \quad \lim_{h \rightarrow 0} \frac{h(4x + 2h + 1)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 4x + 2(0) = 4x$$

Question 5 of 9

Find the following limit.

If the limit does not exist, write "Does Not Exist."

$$\lim_{h \rightarrow 0} \frac{(-1+h)^2 - 1}{h} \quad \lim_{h \rightarrow 0} \frac{(-1+h)(-1+h) - 1}{h} \quad \lim_{h \rightarrow 0} \frac{(1+(-2h)+h^2) - 1}{h} \quad \lim_{h \rightarrow 0} \frac{1 - 2h + h^2 - 1}{h}$$

bottom $\neq 0$

Question 6 of 9

Find the following limit.

If the limit does not exist, write "Does Not Exist."

$$\lim_{x \rightarrow 2} \frac{3x^2 + 4x - 7}{2x + 5} \quad \lim_{x \rightarrow 2} \frac{3(2)^2 + 4(2) - 7}{2(2) + 5} \quad \lim_{x \rightarrow 2} \frac{12 + 8 - 7}{4 + 5}$$

$$\lim_{h \rightarrow 0} -2 + h = -2 + 0 = -2$$

$\downarrow$ I feel like this is more typical

$$\lim_{x \rightarrow 2} \frac{13}{9}$$

$\downarrow$ Simplest Form

$$\lim_{x \rightarrow 2} 1.4$$

Question 7 of 9

Find the average rate of change of $g(x) = -2x^3 + 5x^2$ from $x = 1$ to $x = 2$.

Simplify your answer as much as possible.

$$f(a) = f(1) = -2(1)^3 + 5(1)^2$$

$$f(a) = f(1) = -2 + 5 = 3$$

$$f(a) = f(1) = 3$$

$$Avg[a, b] = \frac{f(b) - f(a)}{b - a}$$

$$f(b) = f(2) = -2(2)^3 + 5(2)^2$$

$$f(b) = f(2) = -2(8) + 5(4)$$

$$f(b) = f(2) = -16 + 20$$

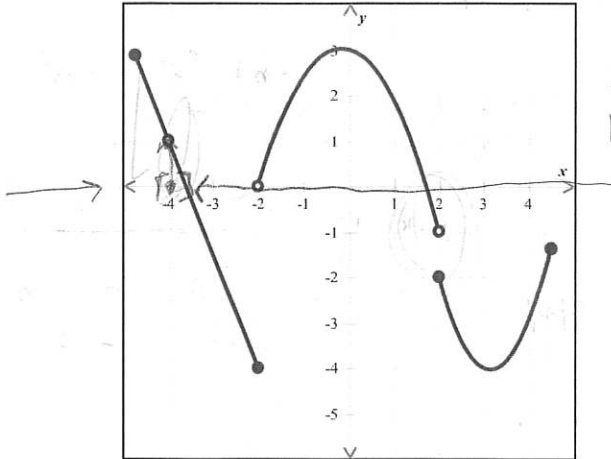
$$Avg[a, b] = \frac{4 - 3}{2 - 1}$$

$$Avg[a, b] = \frac{1}{1} = 1$$

$$f(b) = f(2) = 4$$

Question 8 of 9

The function f is graphed below.



* limits don't give a shit *

originally I thought undefined with $\lim_{x \rightarrow -4^-}$ and $\lim_{x \rightarrow -4^+}$ because of the whole but * limits don't give a shit *

Find the following limits. If a limit does not exist, write "Does Not Exist".

$$\lim_{x \rightarrow -4^-} f(x) = \underline{1}$$

$$\lim_{x \rightarrow -4^+} f(x) = \underline{1}$$

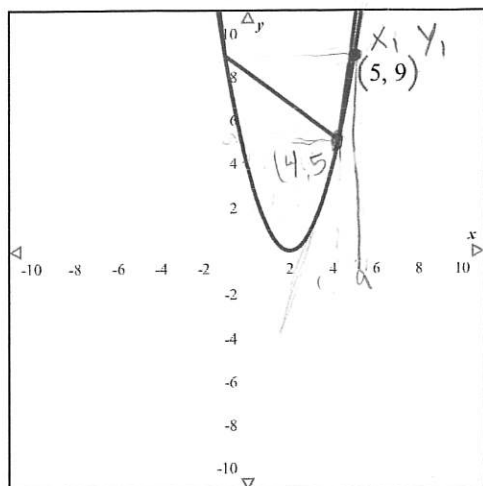
$$\lim_{x \rightarrow -4} f(x) = \underline{1}$$

So 1. I remembered even though -2 has a whole at limit it still has a limit $x \rightarrow -2^+$ but the function is undefined.

* So these have a limit but the function is undefined with hole

Question 9 of 9

The point $(5, 9)$ is on the graph of the function $f(x) = (x-2)^2$ as shown.



(a) Write the equation for the slope of the secant line between $(5, 9)$ and $(a, f(a))$ for the given function $f(x)$ above.

$$x_1 = 5$$

$$m = \frac{f(b) - f(a)}{b - a} = \frac{49 - 9}{9 - 5} = \frac{40}{4} = 10$$

$$y_1 = 9$$

$$f(5) = (5-2)^2 = 9$$

$$EQ: y = m(x - x_1) - y_1$$

$$y = 10(x - 5) - 9$$

+2 looks but I don't know how to get it!

Value of x	4.8	4.9	4.99	4.999	$\rightarrow 5 \leftarrow$	5.001	5.01	5.1	5.2
Slope	5.8	5.9	5.99	5.999	$\rightarrow ? \leftarrow$	6.001	6.01	6.1	6.2

(b) Given the graph and the table above, give the apparent slope of the tangent line to the graph of $f(x)$ at $(5, 9)$.

6

The values of x get put into $f(x) = (x-2)^2 = 9$

x get put into $m = \frac{f(x) - f(5)}{x - 5}$

$$m = \frac{f(x) - f(5)}{x - 5}$$

$$m = \frac{((x-2)^2 - 9)}{x - 5}$$

$$Ex: \frac{((4.8-2)^2 - 9)}{4.8 - 5}$$

calculator

y window

$$y_1 = (x-2)^2$$

$$y_2 = \frac{((x-2)^2 - 9)}{x - 5}$$

table

x	y ₁	y ₂
4.8	7.84	5.8
4.9	8.41	5.9
4.99	8.9401	5.99
4.999	8.994	5.999
5.001	9.006	6.001
5.01	9.0601	6.01
5.1	9.61	6.1
5.2	10.24	6.2

Repeat for all x values!

Q1.) a.) $x = 4.999$

$$\lim_{x \rightarrow 5} \frac{7(4.999) - 5}{(4.999)^2 - 25}$$

$$\lim_{x \rightarrow 5} \frac{-0.007}{-0.009999}$$

$$\lim_{x \rightarrow 5} 0.70070007$$

Rounded = 0.7001

$x = 5.001$

$$\lim_{x \rightarrow 5} \frac{7(5.001) - 5}{(5.001)^2 - 25}$$

$$\lim_{x \rightarrow 5} \frac{0.007}{0.010001}$$

$$\lim_{x \rightarrow 5} 0.699930007$$

Rounded = 0.6999

b.) 0.7

Q2) a.) $f(x) = 4x^2 - 16$

$$f'(x) = \lim_{h \rightarrow 0} \frac{4(2+h)^2 - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{4(2+h)(2+h) - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{4(4 + 4h + h^2) - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{16 + 16h + 4h^2 - 16}{h}$$

$$\lim_{h \rightarrow 0} \frac{16h + 4h^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{h(16 + 4h)}{h}$$

$$\lim_{h \rightarrow 0} 16 + 4h$$

b.) $f'(2) = \lim_{h \rightarrow 0} 16 + 4(0)$

$$f'(2) = \lim_{h \rightarrow 0} 16$$

$$Q3.) \lim_{x \rightarrow 1} \frac{(1)^2 - 2(1) + 7}{1-4}$$

$$\lim_{x \rightarrow 1} \frac{1 - 2 + 7}{-3}$$

$$\lim_{x \rightarrow 1} \frac{6}{-3}$$

$$\lim_{x \rightarrow 1} -2$$

$$Q4.) f'(x) = \lim_{h \rightarrow 0} \frac{2(x+h)^2 + (x+h) - (2x^2 + x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{2(x+h)(x+h) + (x+h) - (2x^2 + x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{2(x^2 + 2xh + h^2) + (x+h) - (2x^2 + x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{2x^2} + 4xh + 2h^2 + \cancel{x} + h - \cancel{2x^2} - \cancel{x}}{h}$$

$$\lim_{h \rightarrow 0} \frac{4xh + 2h^2 + h}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{h}(4x + 2h)}{\cancel{h}}$$

$$\lim_{h \rightarrow 0} 4x + 2h$$

$$b.) f'(x) = \lim_{h \rightarrow 0} 4x + 2(0)$$

$$f'(x) = \lim_{h \rightarrow 0} 4x$$

Q5.) $\lim_{h \rightarrow 0} \frac{(-1+h)^2 - 1}{h}$

$$\lim_{h \rightarrow 0} \frac{(-1+h)(-1+h) - 1}{h}$$

$$\lim_{h \rightarrow 0} \frac{(1 + (-2h) + h^2) - 1}{h}$$

$$\lim_{h \rightarrow 0} \frac{1 - 2h + h^2 - 1}{h}$$

$$\lim_{h \rightarrow 0} \frac{-2h + h^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{-2 + h}{1}$$

$$\lim_{h \rightarrow 0} -2 + h = -2 + 0 = -2$$

$$\lim_{h \rightarrow 0} -2$$

Q6.) $\lim_{x \rightarrow 2} \frac{3(x)^2 + 4x - 7}{2x + 5}$

$$\lim_{x \rightarrow 2} \frac{3(2)^2 + 4(2) - 7}{2(2) + 5}$$

$$\lim_{x \rightarrow 2} \frac{12 + 8 - 7}{4 + 5}$$

$$\lim_{x \rightarrow 2} \frac{13}{9}$$

* more simplified

$$\lim_{x \rightarrow 2} 1.4$$

Q7.) $g(x) = -2x^3 + 5x^2$

$x=1$
 a

$x=2$
 b

$$AV[a, b] = \frac{f(b) - f(a)}{b - a}$$

$$AV[a, b] = \frac{4 - 3}{2 - 1}$$

$$AV[a, b] = 1/1 = 1$$

$$f(a) = f(1) = -2(1)^3 + 5(1)^2$$

$$f(a) = f(1) = -2 + 5$$

$$f(a) = f(1) = 3$$

$$f(b) = f(2) = -2(2)^3 + 5(2)^2$$

$$f(b) = f(2) = -16 + 20$$

$$f(b) = f(2) = 4$$

Q8.) ON TEST

Q9.) ON TEST

$$\frac{(-1+h)(-1+h) - 1}{h} = \frac{(1 + (-1h) + (-1h) + h^2) - 1}{h}$$

$$\frac{(1 + (-2h) + h^2) - 1}{h} = \frac{1 - 2h + h^2 - 1}{h}$$

$$\frac{h(-2+h)}{h} = -2+h$$

$$-2+0 = -2$$

$$\frac{2(x+h)^2 + (x+h) - (2x^2 + x)}{h}$$

$$\frac{2(x+h)(x+h) + (x+h) - (2x^2 + x)}{h}$$

$$\frac{2(x^2 + 2xh + h^2) + (x+h) - (2x^2 + x)}{h}$$

$$\cancel{2x^2} + 4xh + 2h^2 + \cancel{x+h} - \cancel{2x^2} - \cancel{x}$$

$$\frac{4xh + 2h^2 + h}{h}$$

no like terms so
pull h out!

$$\frac{h(4x + 2h)}{h}$$

$$4x + 2h$$

$$4x + 2(0) = 4x$$

